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SECTION:P5
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C PROGRAMS WEEK-10

PROGRAM-1

CODE

```c
//PES1UG25AM095
#include<stdio.h>
#include<string.h>

struct supermarket_sim
{
    int code;
    char name[20];
    int quantity;
    float ppu;
    float discount;
};

int main()
{
    struct supermarket_sim s1;

    printf("Enter item code: ");
    scanf("%d", &s1.code);

    printf("Enter item name: ");
    getchar();
    scanf("%[^\n]", s1.name);

    printf("Enter quantity: ");
    scanf("%d", &s1.quantity);

    printf("Enter price per unit: ");
    scanf("%f", &s1.ppu);

    printf("Enter discount percentage: ");
    scanf("%f", &s1.discount);

    printf("\n---- Item Details -----\n");
    printf("Item Code: %d\n", s1.code);
    printf("Item Name: %s\n", s1.name);
    printf("Quantity: %d\n", s1.quantity);
    printf("Price Per Unit: %.2f\n", s1.ppu);
    printf("Discount Percentage: %.2f\n", s1.discount);

    float total_cost = s1.ppu * s1.quantity;
    float discount_amt = total_cost * (s1.discount / 100);
    float final_amt = total_cost - discount_amt;

    printf("\n--- Bill Details -----\n");
    printf("Total Cost: %.2f\n", total_cost);
    printf("Discount Amount: %.2f\n", discount_amt);
    printf("Final Bill Amount: %.2f\n", final_amt);

    return 0;
```

OUTPUT

```
PS C:\Users\Chethan S\Documents\WEEK-10> cd "c:\
Enter item code: 102
Enter item name: Cooking Oil
Enter quantity: 4
Enter price per unit: 180
Enter discount percentage: 10

---- Item Details -----
Item Code: 102
Item Name: Cooking Oil
Quantity: 4
Price Per Unit: 180.00
Discount Percentage: 10.00

--- Bill Details -----
Total Cost: 720.00
Discount Amount: 72.00
Final Bill Amount: 648.00
PS C:\Users\Chethan S\Documents\WEEK-10>
```

CODE

```c
//PES1UG25AM095
#include<stdio.h>
#include<string.h>

struct patient
{
    int id;
    char name[20];
    int age;
    char disease[20];
    int days_admit;
    int tcpd;
};

int main()
{
    struct patient p1;

    printf("Enter patient ID: ");
    scanf("%d", &p1.id);

    printf("Enter patient name: ");
    getchar();
    scanf("%[^\n]", p1.name);

    printf("Enter patient age: ");
    scanf("%d", &p1.age);

    printf("Enter patient Disease: ");
    getchar();
    scanf("%[^\n]", p1.disease);

    printf("Enter number of days admitted: ");
    scanf("%d", &p1.days_admit);

    printf("Enter treatment charge per day: ");
    scanf("%d", &p1.tcpd);

    printf("\n--- Patient Details ---\n");
    printf("Patient ID: %d\n", p1.id);
    printf("Patient Name: %s\n", p1.name);
    printf("Patient Age: %d\n", p1.age);
    printf("Patient Disease Name: %s\n", p1.disease);
    printf("Days Admitted: %d\n", p1.days_admit);
    printf("Treatment Charge per Day: %d\n", p1.tcpd);

    float total_cost = p1.tcpd * p1.days_admit;
    float discount_amt = 0;
    float extra_charge = 0;

    if(p1.age > 60)
        discount_amt = total_cost * 0.1;

    if(p1.days_admit > 7)
        extra_charge = 2000;

    float final_amt = total_cost + extra_charge - discount_amt;

    printf("\n--- Bill Details ---\n");
    printf("Total Treatment Cost: %.2f\n", total_cost);
    printf("Discount Amount: %.2f\n", discount_amt);
    printf("Extra Medical Charges: %.2f\n", extra_charge);
    printf("Final Hospital Bill: %.2f\n", final_amt);
```

PROGRAM-2

OUTPUT

```
Enter patient ID: 101
Enter patient name: Ramesh Kumar
Enter patient age: 65
Enter patient Disease: Viral Fever
Enter number of days admitted: 10
Enter treatment charge per day: 3000

--- Patient Details ---
Patient ID: 101
Patient Name: Ramesh Kumar
Patient Age: 65
Patient Disease Name: Viral Fever
Days Admitted: 10
Treatment Charge per Day: 3000

--- Bill Details ---
Total Treatment Cost: 30000.00
Discount Amount: 3000.00
Extra Medical Charges: 2000.00
Final Hospital Bill: 29000.00
PS C:\Users\Chethan S\Documents\WEEK-10>
```

CODE

PROGRAM-3

```c
//PES1UG25AM095
#include<stdio.h>
#include<string.h>

struct cricket_players
{
    char name[20];
    int matches;
    int runs;
    int not_out;

    float batting_avg;
};

int main()
{
    int top_player_index = -1;
    float highest_avg = 0;
    int n;

    printf("Enter the number of players: ");
    scanf("%d", &n);

    struct cricket_players c1[n];

    for(int i = 0; i < n; i++)
    {
        printf("Enter details of player %d\n", i + 1);

        printf("Enter player name: ");
        getchar();
        scanf("%[^\n]", c1[i].name);

        printf("Enter matches played: ");
        scanf("%d", &c1[i].matches);

        printf("Enter total runs scored: ");
        scanf("%d", &c1[i].runs);

        printf("Enter number of times not out: ");
        scanf("%d", &c1[i].not_out);

        int denominator = 0;
        denominator = c1[i].matches - c1[i].not_out;

        if(denominator == 0)
            c1[i].batting_avg = 0;
        else
            c1[i].batting_avg = (float)c1[i].runs / denominator;

        if(c1[i].batting_avg > highest_avg)
        {
            highest_avg = c1[i].batting_avg;
            top_player_index = i;
        }
    }
    printf("\n--- Player with Highest Batting Average ---\n");

    if(top_player_index != -1)
    {
        printf("Player Name: %s\n", c1[top_player_index].name);
        printf("Matches Played: %d\n", c1[top_player_index].matches);
        printf("Total Runs: %d\n", c1[top_player_index].runs);
        printf("Not Out: %d\n", c1[top_player_index].not_out);
        printf("Batting Average: %.2f\n", c1[top_player_index].batting_avg);
    }
    return 0;
}
```

OUTPUT

```
($?) { .\c3 }
Enter the number of players: 3
Enter details of player 1
Enter player name: Virat Kohli
Enter matches played: 100
Enter total runs scored: 5000
Enter number of times not out: 10
Enter details of player 2
Enter player name: Rohit Sharma
Enter matches played: 95
Enter total runs scored: 4800
Enter number of times not out: 5
Enter details of player 3
Enter player name: Shreyas Iyer
Enter matches played: 80
Enter total runs scored: 3900
Enter number of times not out: 20

--- Player with Highest Batting Average ---
Player Name: Shreyas Iyer
Matches Played: 80
Total Runs: 3900
Not Out: 20
Batting Average: 65.00
PS C:\Users\Chethan S\Documents\WEEK-10>
```

PROGRAM-4

CODE

```c
//PES1UG25AM095
#include<stdio.h>
#include<string.h>

struct students
{
    char usn[10];
    char name[10];
    int marks[3];
    float attendance;

    float scholarship;
    char grade;
    char eligibility[20];
    char result[10];
    float avg;
    int total;
};

int main()
{
    int n, top_student_index = -1;
    int pass = 0, fail = 0, not_eligible = 0;
    float highest_avg = 0;

    printf("Enter number of students: ");
    scanf("%d", &n);

    struct students s1[n];

    for(int i = 0; i < n; i++)
    {
        printf("Enter details of student %d\n", i + 1);

        printf("Enter USN: ");
        getchar();
        scanf("%[^\n]", s1[i].usn);

        printf("Enter Name: ");
        getchar();
        scanf("%[^\n]", s1[i].name);

        s1[i].total = 0;

        printf("Enter marks in 3 subjects: ");
        for(int j = 0; j < 3; j++)
        {
            scanf("%d", &s1[i].marks[j]);
            s1[i].total += s1[i].marks[j];
        }

        printf("Enter attendance percentage: ");
        scanf("%f", &s1[i].attendance);

        s1[i].avg = s1[i].total / 3.0;

        if(s1[i].marks[0] < 35 || s1[i].marks[1] < 35 || s1[i].marks[2] < 35)
        {
            strcpy(s1[i].result, "Fail");
            fail++;
        }
        else
        {
            strcpy(s1[i].result, "Pass");
            pass++;
        }

        if(s1[i].avg >= 85)
        {
            s1[i].grade = 'A';
        }
        else if(s1[i].avg >= 70 && s1[i].avg < 85)
        {
            s1[i].grade = 'B';
        }
        else if(s1[i].avg >= 50 && s1[i].avg < 70)
        {
            s1[i].grade = 'C';
        }
        else if(s1[i].avg < 50)
        {
            s1[i].grade = 'D';
        }

        if(s1[i].attendance < 75)
        {
            strcpy(s1[i].eligibility, "Not Eligible");
            not_eligible++;
        }
        else
        {
            strcpy(s1[i].eligibility, "Eligible");
        }

        s1[i].scholarship = 0;

        if(strcmp(s1[i].result, "Pass") == 0 && strcmp(s1[i].eligibility, "Eligible") == 0)
        {
            if(s1[i].avg >= 90 && s1[i].attendance >= 90)
                s1[i].scholarship = 10000;
            else if(s1[i].avg >= 80 && s1[i].attendance >= 85)
                s1[i].scholarship = 5000;
            else
                s1[i].scholarship = 0;
        }

        if(s1[i].avg > highest_avg && strcmp(s1[i].eligibility, "Eligible") == 0)
        {
            highest_avg = s1[i].avg;
            top_student_index = i;
        }
    }

    for(int i = 0; i < n; i++)
    {
        printf("\nStudent %d\n", i + 1);
        printf("USN: %s\n", s1[i].usn);
        printf("Name: %s\n", s1[i].name);

        printf("Marks: ");
        for(int j = 0; j < 3; j++)
            printf("%d ", s1[i].marks[j]);

        printf("\nAttendance: %.2f\n", s1[i].attendance);
        printf("Total: %d\n", s1[i].total);
        printf("Average: %.2f\n", s1[i].avg);
        printf("Result: %s\n", s1[i].result);
        printf("Grade: %c\n", s1[i].grade);
        printf("Eligibility: %s\n", s1[i].eligibility);
        printf("Scholarship: %.2f\n", s1[i].scholarship);
    }

    printf("\n========= Summary =========\n");
    printf("Passed Students : %d\n", pass);
    printf("Failed Students : %d\n", fail);
    printf("Students Not Eligible by Attendance : %d\n", not_eligible);

    printf("\n===== Top Eligible Student =====\n");
    if(top_student_index != -1)
    {
        printf("USN: %s\n", s1[top_student_index].usn);
        printf("Name: %s\n", s1[top_student_index].name);
        printf("Average: %.2f\n", s1[top_student_index].avg);
        printf("Attendance: %.2f\n", s1[top_student_index].attendance);
    }

    return 0;
}
```

OUTPUT

```
PS C:\Users\Chethan S\Documents\WEEK-10> cd "c:\Us
Enter number of students: 3


Enter details of student 1
Enter USN: PES001
Enter Name: Amit Kumar
Enter marks in 3 subjects: 91 88 95
Enter attendance percentage: 92


Enter details of student 2
Enter USN: PES002
Enter Name: Sneha Rao
Enter marks in 3 subjects: 78 69 72
Enter attendance percentage: 82


Enter details of student 3
Enter USN: PES003
Enter Name: Rahul Sharma
Enter marks in 3 subjects: 30 67 80
Enter attendance percentage: 70

Student 1
USN: PES001
Name: Amit Kumar[
Marks: 91 88 95
Attendance: 92.00
Total: 274
Average: 91.33
Result: Pass
Grade: A
Eligibility: Eligible
Scholarship: 10000.00

Student 2
USN: PES002
Name: Sneha Rao
Marks: 78 69 72
Attendance: 82.00
Total: 219
Average: 73.00
Result: Pass
Grade: B
Eligibility: Eligible
Scholarship: 0.00

Student 3
USN: PES003
Name: Rahul Shar
Marks: 30 67 80
Attendance: 70.00
Total: 177
Average: 59.00
Result: Fail
Grade: C
Eligibility: Not Eligible
Scholarship: 0.00

========= Summary =========
Passed Students : 2
Failed Students : 1
Students Not Eligible by Attendance : 1

===== Top Eligible Student =====
USN: PES001
Name: Amit Kumar[
Average: 91.33
Attendance: 92.00
PS C:\Users\Chethan S\Documents\WEEK-10>
```